data-cleaning-task

March 26, 2026

Getting the Data from MYSQL

[2]:
```
!pip install sqlalchemy
```

```
Requirement already satisfied: sqlalchemy in d:\programdata\lib\site-packages
(2.0.39)
Requirement already satisfied: greenlet!=0.4.17 in d:\programdata\lib\site-
packages (from sqlalchemy) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in
d:\programdata\lib\site-packages (from sqlalchemy) (4.15.0)
```

[1]:
```
from sqlalchemy import create_engine
import pandas as pd
import warnings
warnings.filterwarnings('ignore')
engine= create_engine("mysql+mysqlconnector://root:050620@localhost:3306/
 ↪transactions")
df=pd.read_sql("select * from upi_transactions",engine)
df.head()
```

[1]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      02/02/2024   271.64     SBI Bank        HDFC Bank
1      TXN00002      03/03/2024  1064.63   ICICI Bank         SBI Bank
2      TXN00003      04/04/2024   144.15    Axis Bank        Axis Bank
3      TXN00004      05/05/2024   612.89    HDFC Bank       ICICI Bank
4      TXN00005      06/06/2024   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0     05:12:14 PM     Tablet  Phone Number       Amazon          Food
1     11:15:02 AM     Laptop       QR Code       Zomato        Travel
2     09:29:39 PM     Mobile        UPI ID       Swiggy  Bill Payment
3     06:27:36 AM     Tablet  Phone Number        IRCTC        Others
```

```
4     02:06:22 AM     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
3           24     Instant      INR           123456789016
4           25   Scheduled      USD           123456789017

   MerchantAccountNumber
0           987654321013
1           987654321014
2           987654321015
3           987654321016
4           987654321017
```

[2]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20185 entries, 0 to 20184
Data columns (total 20 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   TransactionID          20185 non-null  object
 1   TransactionDate        20185 non-null  object
 2   Amount                 20185 non-null  float64
 3   BankNameSent           20185 non-null  object
 4   BankNameReceived       20185 non-null  object
 5   RemainingBalance       20185 non-null  float64
 6   City                   20185 non-null  object
 7   Gender                 20185 non-null  object
 8   TransactionType        20185 non-null  object
 9   Status                 20185 non-null  object
 10  TransactionTime        20185 non-null  object
 11  DeviceType             20185 non-null  object
 12  PaymentMethod          20185 non-null  object
 13  MerchantName           20185 non-null  object
 14  Purpose                20185 non-null  object
 15  CustomerAge            20185 non-null  int64
 16  PaymentMode            20185 non-null  object
 17  Currency               20185 non-null  object
 18  CustomerAccountNumber  20185 non-null  int64
 19  MerchantAccountNumber  20185 non-null  int64
dtypes: float64(2), int64(3), object(15)
memory usage: 3.1+ MB
```

[3]:
```
df.describe()
```

[3]:
```
              Amount  RemainingBalance   CustomerAge  CustomerAccountNumber  \
count  20185.000000      20185.000000  20185.000000           2.018500e+04
mean     992.632673       5019.457372     39.509091           1.234568e+11
std      577.237229       2893.550725     11.604222           5.781609e+03
min     -456.000000          0.530000      0.000000           1.234568e+11
25%      496.770000       2529.360000     29.000000           1.234568e+11
50%      982.830000       5019.350000     39.000000           1.234568e+11
75%     1494.520000       7530.510000     50.000000           1.234568e+11
max     1999.870000       9999.470000     99.000000           1.234568e+11

       MerchantAccountNumber
count           2.018500e+04
mean            9.876543e+11
std             5.781609e+03
min             9.876543e+11
25%             9.876543e+11
50%             9.876543e+11
75%             9.876543e+11
max             9.876543e+11
```

[4]:
```
df.head(2)
```

[4]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      02/02/2024   271.64     SBI Bank        HDFC Bank
1      TXN00002      03/03/2024  1064.63   ICICI Bank         SBI Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success

  TransactionTime DeviceType PaymentMethod MerchantName Purpose  CustomerAge  \
0     05:12:14 PM     Tablet  Phone Number       Amazon    Food           21
1     11:15:02 AM     Laptop       QR Code       Zomato  Travel           22

  PaymentMode Currency  CustomerAccountNumber  MerchantAccountNumber
0   Scheduled      USD           123456789013           987654321013
1     Instant      EUR           123456789014           987654321014
```

1. Remove duplicates TransactionID

[6]:
```
df["TransactionID"].duplicated().sum()
```

[6]:
```
np.int64(185)
```

There are 185 duplicates values in the TransactionID column.

remove duplicates in TransactionID

[7]:
```
## Duplicated rows

df[df["TransactionID"].duplicated()]
```

[7]:
```
     TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
2836      TXN03148      05/29/2024   285.75    HDFC Bank       ICICI Bank
3683      TXN03193      02/14/2024    31.61     SBI Bank        HDFC Bank
3684      TXN03194      03/15/2024   814.30   ICICI Bank         SBI Bank
3685      TXN03195      04/16/2024  1634.05    Axis Bank        Axis Bank
3686      TXN03196      05/17/2024  1183.27    HDFC Bank       ICICI Bank
…            …             …       …          …               …
5158      TXN05021      06/12/2024   921.33     SBI Bank        HDFC Bank
5159      TXN05022      07/13/2024   925.44   ICICI Bank         SBI Bank
5160      TXN05023      08/14/2024  1653.41    Axis Bank        Axis Bank
5161      TXN05024      09/15/2024  1890.44    HDFC Bank       ICICI Bank
5162      TXN05025      10/16/2024  1155.43     SBI Bank        HDFC Bank

      RemainingBalance       City  Gender TransactionType   Status  \
2836           6614.63     Mumbai    Male         Payment  Success
3683           8227.77      Delhi  Female        Transfer  Success
3684           3797.33  Bangalore    Male         Payment  Success
3685           3238.90  Hyderabad  Female        Transfer   Failed
3686           5743.51     Mumbai    Male         Payment  Success
…                …        …     …              …      …
5158            769.22      Delhi  Female        Transfer  Success
5159           7034.73  Bangalore    Male         Payment  Success
5160           5450.36  Hyderabad  Female        Transfer  Success
5161           7425.34     Mumbai    Male         Payment  Success
5162           3817.08      Delhi  Female        Transfer   Failed

     TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
2836     12:29:58 PM     Tablet  Phone Number       Swiggy  Bill Payment
3683     11:12:55 AM     Tablet  Phone Number       Swiggy  Bill Payment
3684     07:35:51 PM     Laptop       QR Code        IRCTC        Others
3685     01:20:36 AM     Mobile        UPI ID     Flipkart      Shopping
3686     07:50:57 PM     Tablet  Phone Number       Amazon          Food
…              …        …            …          …            …
5158     08:18:32 PM     Laptop       QR Code       Amazon          Food
5159     08:57:08 PM     Mobile        UPI ID       Zomato        Travel
5160     11:18:12 PM     Tablet  Phone Number       Swiggy  Bill Payment
5161     06:38:05 PM     Laptop       QR Code        IRCTC        Others
5162     05:49:45 PM     Mobile        UPI ID     Flipkart      Shopping

      CustomerAge PaymentMode Currency  CustomerAccountNumber  \
2836           48     Instant      INR           123456792160
3683           53   Scheduled      USD           123456792205
3684           54     Instant      EUR           123456792206
```

```
3685            55   Scheduled      GBP            123456792207
3686            56     Instant      INR            123456792208
…            …          …       …                    …
5158            41   Scheduled      USD            123456794033
5159            42     Instant      EUR            123456794034
5160            43   Scheduled      GBP            123456794035
5161            44     Instant      INR            123456794036
5162            45   Scheduled      USD            123456794037

      MerchantAccountNumber
2836           987654324160
3683           987654324205
3684           987654324206
3685           987654324207
3686           987654324208
…                     …
5158           987654326033
5159           987654326034
5160           987654326035
5161           987654326036
5162           987654326037

[185 rows x 20 columns]
```

[8]:
```python
## Removing Duplicates
df.drop_duplicates(subset="TransactionID",keep="first",inplace=True)
```

[9]:
```python
df["TransactionID"].duplicated().sum()
```

[9]:
```
np.int64(0)
```

From above All duplicate TransactionID values have been successfully removed

[10]:
```python
df.head()
```

[10]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      02/02/2024   271.64     SBI Bank        HDFC Bank
1      TXN00002      03/03/2024  1064.63   ICICI Bank         SBI Bank
2      TXN00003      04/04/2024   144.15    Axis Bank        Axis Bank
3      TXN00004      05/05/2024   612.89    HDFC Bank       ICICI Bank
4      TXN00005      06/06/2024   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
```

```
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0     05:12:14 PM     Tablet  Phone Number       Amazon          Food
1     11:15:02 AM     Laptop       QR Code       Zomato        Travel
2     09:29:39 PM     Mobile        UPI ID       Swiggy  Bill Payment
3     06:27:36 AM     Tablet  Phone Number        IRCTC        Others
4     02:06:22 AM     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
3           24     Instant      INR           123456789016
4           25   Scheduled      USD           123456789017

   MerchantAccountNumber
0           987654321013
1           987654321014
2           987654321015
3           987654321016
4           987654321017
```

2. Standardize Gender values

[11]:
```
df["Gender"].unique()
```

[11]:
```
array(['Female', 'Male', 'M', 'm', 'f', 'male', 'female', 'F'],
      dtype=object)
```

[12]:
```
## Converting to lower case
df["Gender"]=df["Gender"].str.lower()
```

[13]:
```
df["Gender"].unique()
```

[13]:
```
array(['female', 'male', 'm', 'f'], dtype=object)
```

[14]:
```
## Standardize Gender Values
df["Gender"]=df["Gender"].map({"female":"Female","male":"Male","m":"Male","f":
 ↪"Female"})
```

[16]:
```
df["Gender"].unique()
```

[16]:
```
array(['Female', 'Male'], dtype=object)
```

Finally successfully standardized all gender values into only two consistent categories: "Male" and "Female".

[17]:
```
df
```

[17]:
```
       TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0           TXN00001      02/02/2024   271.64     SBI Bank        HDFC Bank
1           TXN00002      03/03/2024  1064.63   ICICI Bank         SBI Bank
2           TXN00003      04/04/2024   144.15    Axis Bank        Axis Bank
3           TXN00004      05/05/2024   612.89    HDFC Bank       ICICI Bank
4           TXN00005      06/06/2024   743.32     SBI Bank        HDFC Bank
…               …             …       …           …                …
20180       TXN00619      08/20/2024  1557.63    Axis Bank        Axis Bank
20181       TXN00620      09/21/2024  1151.00    HDFC Bank       ICICI Bank
20182       TXN00621      10/22/2024  1928.98     SBI Bank        HDFC Bank
20183       TXN00622      11/23/2024   929.87   ICICI Bank         SBI Bank
20184       TXN00623      12/24/2024   702.63    Axis Bank        Axis Bank

        RemainingBalance       City  Gender TransactionType   Status  \
0                5557.02      Delhi  Female        Transfer  Success
1                9753.32  Bangalore    Male         Payment  Success
2                7597.35  Hyderabad  Female        Transfer  Success
3                2327.84     Mumbai    Male         Payment  Success
4                1136.84      Delhi  Female        Transfer   Failed
…                  …         …       …              …        …
20180            5379.78  Hyderabad  Female        Transfer  Success
20181            8024.64     Mumbai    Male         Payment   Failed
20182            9428.43      Delhi  Female        Transfer  Success
20183            2858.71  Bangalore    Male         Payment  Success
20184            8392.23  Hyderabad  Female        Transfer  Success

       TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0          05:12:14 PM     Tablet  Phone Number       Amazon          Food
1          11:15:02 AM     Laptop       QR Code       Zomato        Travel
2          09:29:39 PM     Mobile        UPI ID       Swiggy  Bill Payment
3          06:27:36 AM     Tablet  Phone Number        IRCTC        Others
4          02:06:22 AM     Laptop       QR Code     Flipkart      Shopping
…                …          …            …           …            …
20180      06:27:06 PM     Tablet  Phone Number                     Others
20181      10:04:29 PM     Laptop       QR Code     Flipkart      Shopping
20182      05:19:18 AM     Mobile        UPI ID       Amazon          Food
20183      07:43:50 PM     Tablet  Phone Number       Zomato        Travel
20184      12:30:33 AM     Laptop       QR Code       Swiggy  Bill Payment

        CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0                21   Scheduled      USD           123456789013
1                22     Instant      EUR           123456789014
2                23   Scheduled      GBP           123456789015
3                24     Instant      INR           123456789016
4                25   Scheduled      USD           123456789017
```

```
…              …          …       …                        …
20180           39   Scheduled      GBP            123456789631
20181           40     Instant      INR            123456789632
20182           41   Scheduled      USD            123456789633
20183           42     Instant      EUR            123456789634
20184           43   Scheduled      GBP            123456789635

       MerchantAccountNumber
0               987654321013
1               987654321014
2               987654321015
3               987654321016
4               987654321017
…                        …
20180           987654321631
20181           987654321632
20182           987654321633
20183           987654321634
20184           987654321635

[20000 rows x 20 columns]
```

3. Convert TransactionDate & TransactionTime properly

[18]:
```
df["TransactionDate"]=pd.to_datetime(df["TransactionDate"],format="%m/%d/%Y").
↪dt.date
```

[19]:
```
df.head()
```

[19]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      2024-02-02   271.64     SBI Bank        HDFC Bank
1      TXN00002      2024-03-03  1064.63   ICICI Bank         SBI Bank
2      TXN00003      2024-04-04   144.15    Axis Bank        Axis Bank
3      TXN00004      2024-05-05   612.89    HDFC Bank       ICICI Bank
4      TXN00005      2024-06-06   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0     05:12:14 PM     Tablet  Phone Number       Amazon          Food
1     11:15:02 AM     Laptop       QR Code       Zomato        Travel
2     09:29:39 PM     Mobile        UPI ID       Swiggy  Bill Payment
```

```
3     06:27:36 AM     Tablet  Phone Number        IRCTC        Others
4     02:06:22 AM     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
3           24     Instant      INR           123456789016
4           25   Scheduled      USD           123456789017

   MerchantAccountNumber
0           987654321013
1           987654321014
2           987654321015
3           987654321016
4           987654321017
```

[20]:
```
df["TransactionTime"]=pd.to_datetime(df["TransactionTime"],format="%I:%M:%S␣
↪%p").dt.time
```

[21]:
```
df.head()
```

[21]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      2024-02-02   271.64     SBI Bank        HDFC Bank
1      TXN00002      2024-03-03  1064.63   ICICI Bank         SBI Bank
2      TXN00003      2024-04-04   144.15    Axis Bank        Axis Bank
3      TXN00004      2024-05-05   612.89    HDFC Bank       ICICI Bank
4      TXN00005      2024-06-06   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0        17:12:14     Tablet  Phone Number       Amazon          Food
1        11:15:02     Laptop       QR Code       Zomato        Travel
2        21:29:39     Mobile        UPI ID       Swiggy  Bill Payment
3        06:27:36     Tablet  Phone Number        IRCTC        Others
4        02:06:22     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
```

```
3            24     Instant      INR            123456789016
4            25   Scheduled      USD            123456789017

   MerchantAccountNumber
0            987654321013
1            987654321014
2            987654321015
3            987654321016
4            987654321017
```

TransactionDate and TransactionTime were successfully converted into proper date-time formats, ensuring accurate time-based analysis.

4. Handle missing MerchantName

[19]:
```
df["MerchantName"].unique()
```

[19]:
```
array(['Amazon', 'Zomato', 'Swiggy', 'IRCTC', 'Flipkart', ''],
      dtype=object)
```

[20]:
```
df["MerchantName"].value_counts()
```

[20]:
```
MerchantName
Amazon      4000
Zomato      4000
Swiggy      4000
Flipkart    4000
IRCTC       3993
               7
Name: count, dtype: int64
```

From above data 7 Blanks are present in the dataset

[21]:
```
df["MerchantName"]=df["MerchantName"].replace('','IRCTC')
```

[22]:
```
df["MerchantName"].unique()
```

[22]:
```
array(['Amazon', 'Zomato', 'Swiggy', 'IRCTC', 'Flipkart'], dtype=object)
```

[23]:
```
df["MerchantName"].value_counts()
```

[23]:
```
MerchantName
Amazon      4000
Zomato      4000
Swiggy      4000
IRCTC       4000
Flipkart    4000
Name: count, dtype: int64
```

Finally making the column complete with no blanks

[24]:
```
df.head()
```

[24]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      02/02/2024   271.64     SBI Bank        HDFC Bank
1      TXN00002      03/03/2024  1064.63   ICICI Bank         SBI Bank
2      TXN00003      04/04/2024   144.15    Axis Bank        Axis Bank
3      TXN00004      05/05/2024   612.89    HDFC Bank       ICICI Bank
4      TXN00005      06/06/2024   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0     05:12:14 PM     Tablet  Phone Number       Amazon          Food
1     11:15:02 AM     Laptop       QR Code       Zomato        Travel
2     09:29:39 PM     Mobile        UPI ID       Swiggy  Bill Payment
3     06:27:36 AM     Tablet  Phone Number        IRCTC        Others
4     02:06:22 AM     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
3           24     Instant      INR           123456789016
4           25   Scheduled      USD           123456789017

   MerchantAccountNumber
0           987654321013
1           987654321014
2           987654321015
3           987654321016
4           987654321017
```

5. Validate negative or zero Amount

[25]:
```
(df["Amount"]<=0).sum()
```

[25]:
```
np.int64(16)
```

There are 16 Transactions where Amount is negative or less than "0"

[26]:
```
df[df["Amount"]<=0]
```

[26]:

	TransactionID	TransactionDate	Amount	BankNameSent	BankNameReceived	\
154	TXN00155	12/06/2024	0.0	Axis Bank	Axis Bank	
431	TXN00744	01/25/2024	-20.0	HDFC Bank	ICICI Bank	
757	TXN01070	03/21/2024	0.0	ICICI Bank	SBI Bank	
913	TXN01226	03/27/2024	-456.0	ICICI Bank	SBI Bank	
1234	TXN01547	12/18/2024	-56.0	Axis Bank	Axis Bank	
1235	TXN01548	01/19/2024	-344.0	HDFC Bank	ICICI Bank	
1236	TXN01549	02/20/2024	0.0	SBI Bank	HDFC Bank	
1237	TXN01550	03/21/2024	0.0	ICICI Bank	SBI Bank	
1238	TXN01551	04/22/2024	-456.0	Axis Bank	Axis Bank	
1239	TXN01552	05/23/2024	0.0	HDFC Bank	ICICI Bank	
1240	TXN01553	06/24/2024	-454.0	SBI Bank	HDFC Bank	
1241	TXN01554	07/25/2024	-46.0	ICICI Bank	SBI Bank	
1242	TXN01555	08/26/2024	0.0	Axis Bank	Axis Bank	
1243	TXN01556	09/27/2024	0.0	HDFC Bank	ICICI Bank	
1552	TXN01865	06/06/2024	0.0	SBI Bank	HDFC Bank	
1680	TXN01993	02/14/2024	-46.0	SBI Bank	HDFC Bank	

	RemainingBalance	City	Gender	TransactionType	Status	\
154	757.89	Hyderabad	Female	Transfer	Failed	
431	434.78	Mumbai	Male	Payment	Success	
757	4540.25	Bangalore	Male	Payment	Failed	
913	3782.99	Bangalore	Male	Payment	Success	
1234	6847.53	Hyderabad	Female	Transfer	Success	
1235	413.90	Mumbai	Male	Payment	Success	
1236	5595.84	Delhi	Female	Transfer	Success	
1237	9572.68	Bangalore	Male	Payment	Failed	
1238	9575.78	Hyderabad	Female	Transfer	Success	
1239	6901.32	Mumbai	Male	Payment	Success	
1240	666.30	Delhi	Female	Transfer	Success	
1241	6021.42	Bangalore	Male	Payment	Success	
1242	5026.40	Hyderabad	Female	Transfer	Failed	
1243	3469.45	Mumbai	Male	Payment	Success	
1552	1759.90	Delhi	Female	Transfer	Failed	
1680	9444.25	Delhi	Female	Transfer	Success	

	TransactionTime	DeviceType	PaymentMethod	MerchantName	Purpose	\
154	08:37:36 AM	Laptop	QR Code	Flipkart	Shopping	
431	02:55:24 AM	Mobile	UPI ID	IRCTC	Others	
757	03:09:47 AM	Laptop	QR Code	Flipkart	Shopping	
913	01:24:58 AM	Laptop	QR Code	Amazon	Food	
1234	11:35:15 AM	Laptop	QR Code	Zomato	Travel	
1235	03:22:24 AM	Mobile	UPI ID	Swiggy	Bill Payment	
1236	12:14:30 AM	Tablet	Phone Number	IRCTC	Others	
1237	04:07:47 AM	Laptop	QR Code	Flipkart	Shopping	
1238	08:12:39 PM	Mobile	UPI ID	Amazon	Food	
1239	01:16:40 PM	Tablet	Phone Number	Zomato	Travel	

```
1240     10:50:21 AM     Laptop       QR Code       Swiggy  Bill Payment
1241     02:01:02 PM     Mobile        UPI ID        IRCTC        Others
1242     02:53:28 AM     Tablet  Phone Number     Flipkart      Shopping
1243     05:05:08 AM     Laptop       QR Code       Amazon          Food
1552     10:16:25 AM     Laptop       QR Code     Flipkart      Shopping
1680     03:20:22 AM     Tablet  Phone Number       Swiggy  Bill Payment

      CustomerAge PaymentMode Currency  CustomerAccountNumber  \
154            55   Scheduled      GBP           123456789167
431            44     Instant      INR           123456789756
757            50     Instant      EUR           123456790082
913            46     Instant      EUR           123456790238
1234           47   Scheduled      GBP           123456790559
1235           48     Instant      INR           123456790560
1236           49   Scheduled      USD           123456790561
1237           50     Instant      EUR           123456790562
1238           51   Scheduled      GBP           123456790563
1239           52     Instant      INR           123456790564
1240           53   Scheduled      USD           123456790565
1241           54     Instant      EUR           123456790566
1242           55   Scheduled      GBP           123456790567
1243           56     Instant      INR           123456790568
1552           45   Scheduled      USD           123456790877
1680           53   Scheduled      USD           123456791005

      MerchantAccountNumber
154            987654321167
431            987654321756
757            987654322082
913            987654322238
1234           987654322559
1235           987654322560
1236           987654322561
1237           987654322562
1238           987654322563
1239           987654322564
1240           987654322565
1241           987654322566
1242           987654322567
1243           987654322568
1552           987654322877
1680           987654323005
```

Replace with Mean amount

[27]:
```
import numpy as np
mean_value=df[df["Amount"]>0]["Amount"].mean()
```

[28]:
```
mean_value
```

[28]: np.float64(993.6361103883107)

[29]:
```
df.loc[df["Amount"]<=0,"Amount"]=np.nan
```

[32]:
```
df["Amount"]=df["Amount"].fillna(mean_value)
```

[33]:
```
(df["Amount"]<=0).sum()
```

[33]: np.int64(0)

There are no more negative or Zero Values in the Data Set

[34]:
```
df.head()
```

[34]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      02/02/2024   271.64     SBI Bank        HDFC Bank
1      TXN00002      03/03/2024  1064.63   ICICI Bank         SBI Bank
2      TXN00003      04/04/2024   144.15    Axis Bank        Axis Bank
3      TXN00004      05/05/2024   612.89    HDFC Bank       ICICI Bank
4      TXN00005      06/06/2024   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0     05:12:14 PM     Tablet  Phone Number       Amazon          Food
1     11:15:02 AM     Laptop       QR Code       Zomato        Travel
2     09:29:39 PM     Mobile        UPI ID       Swiggy  Bill Payment
3     06:27:36 AM     Tablet  Phone Number        IRCTC        Others
4     02:06:22 AM     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
3           24     Instant      INR           123456789016
4           25   Scheduled      USD           123456789017

   MerchantAccountNumber
0           987654321013
1           987654321014
```

```
2            987654321015
3            987654321016
4            987654321017
```

6. Remove invalid ages (<18 or >90)

[35]:
```
df.loc[(df["CustomerAge"]<18) | (df["CustomerAge"]>90),"CustomerAge"]
```

[35]:
```
379      17
614      99
20065     0
20066     9
20067     4
20068     2
20069    95
20070    93
20071    91
20072    95
20073     3
20178    97
20179    99
Name: CustomerAge, dtype: int64
```

[36]:
```
(df.loc[(df["CustomerAge"]<18) | (df["CustomerAge"]>90),"CustomerAge"]).shape[0]
```

[36]:
```
13
```

There are 13 Values are less than 18 and Greater Than 90

[37]:
```
df.drop(df[(df["CustomerAge"]<18) | (df["CustomerAge"]>90)].index,inplace=True)
```

[38]:
```
df.loc[(df["CustomerAge"]<18) | (df["CustomerAge"]>90),"CustomerAge"]
```

[38]:
```
Series([], Name: CustomerAge, dtype: int64)
```

[39]:
```
(df.loc[(df["CustomerAge"]<18) | (df["CustomerAge"]>90),"CustomerAge"]).shape[0]
```

[39]:
```
0
```

[40]:
```
df.shape
```

[40]:
```
(19987, 20)
```

0 Invalid age Records remain

[42]:
```
from sqlalchemy import create_engine
engine= create_engine("mysql+mysqlconnector://root:050620@localhost:3306/
 ↪transactions")
```

```
df.to_sql("Transactions_UPI1",con=engine,if_exists="replace",index=False)
```

[ ]:
```
df.to_csv("Transactions_UPI1.csv",index=False)
```

0.0.1 EDA Activity

Getting the Data From MySQL

[43]:
```
from sqlalchemy import create_engine
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
engine=create_engine("mysql+mysqlconnector://root:050620@localhost:3306/
 ↪transactions")
df=pd.read_sql("select * from transactions_upi",engine)
```

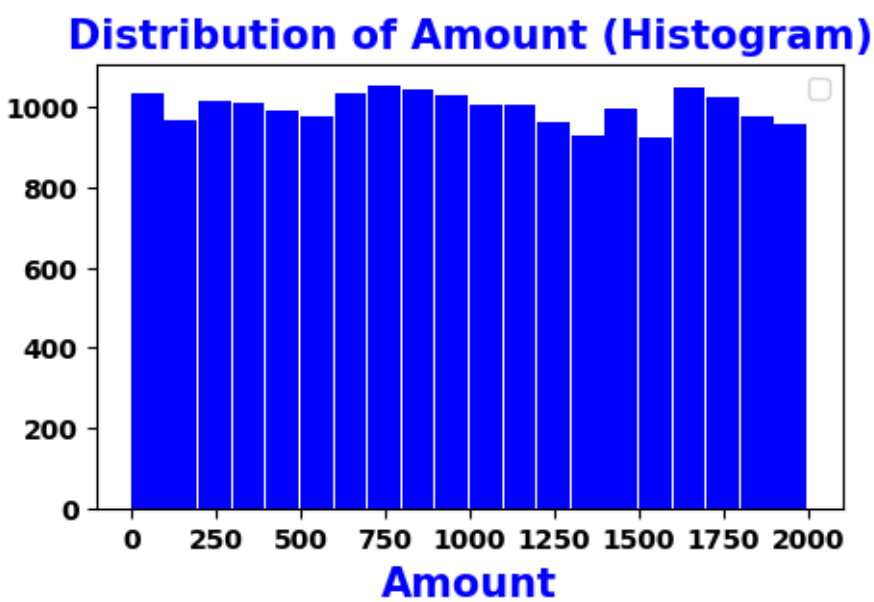
[44]:
```
df["TransactionTime"]=pd.to_datetime(df["TransactionTime"].astype(str).str.
 ↪split().str[-1],format="%H:%M:%S").dt.time
df.head()
```

[44]:
```
  TransactionID TransactionDate   Amount BankNameSent BankNameReceived  \
0      TXN00001      2024-02-02   271.64     SBI Bank        HDFC Bank
1      TXN00002      2024-03-03  1064.63   ICICI Bank         SBI Bank
2      TXN00003      2024-04-04   144.15    Axis Bank        Axis Bank
3      TXN00004      2024-05-05   612.89    HDFC Bank       ICICI Bank
4      TXN00005      2024-06-06   743.32     SBI Bank        HDFC Bank

   RemainingBalance       City  Gender TransactionType   Status  \
0           5557.02      Delhi  Female        Transfer  Success
1           9753.32  Bangalore    Male         Payment  Success
2           7597.35  Hyderabad  Female        Transfer  Success
3           2327.84     Mumbai    Male         Payment  Success
4           1136.84      Delhi  Female        Transfer   Failed

  TransactionTime DeviceType PaymentMethod MerchantName       Purpose  \
0        17:12:14     Tablet  Phone Number       Amazon          Food
1        11:15:02     Laptop       QR Code       Zomato        Travel
2        21:29:39     Mobile        UPI ID       Swiggy  Bill Payment
3        06:27:36     Tablet  Phone Number        IRCTC        Others
4        02:06:22     Laptop       QR Code     Flipkart      Shopping

   CustomerAge PaymentMode Currency  CustomerAccountNumber  \
0           21   Scheduled      USD           123456789013
1           22     Instant      EUR           123456789014
2           23   Scheduled      GBP           123456789015
```

```
3            24     Instant      INR            123456789016
4            25   Scheduled      USD            123456789017

   MerchantAccountNumber
0           987654321013
1           987654321014
2           987654321015
3           987654321016
4           987654321017
```

1. Distribution of Amount (Histogram)

[45]:
```
plt.figure(figsize=(5,3))
plt.hist(df["Amount"],bins=20,width=95,color="blue")
plt.title("Distribution of Amount␣
 ↪(Histogram)",fontweight="bold",fontsize=15,color="blue",loc="center")
plt.xlabel("Amount",fontweight="bold",fontsize=15,color="blue")
plt.xticks(color="Black",fontweight="bold",fontsize=10)
plt.yticks(color="Black",fontweight="bold",fontsize=10)
plt.legend()
plt.show()
```

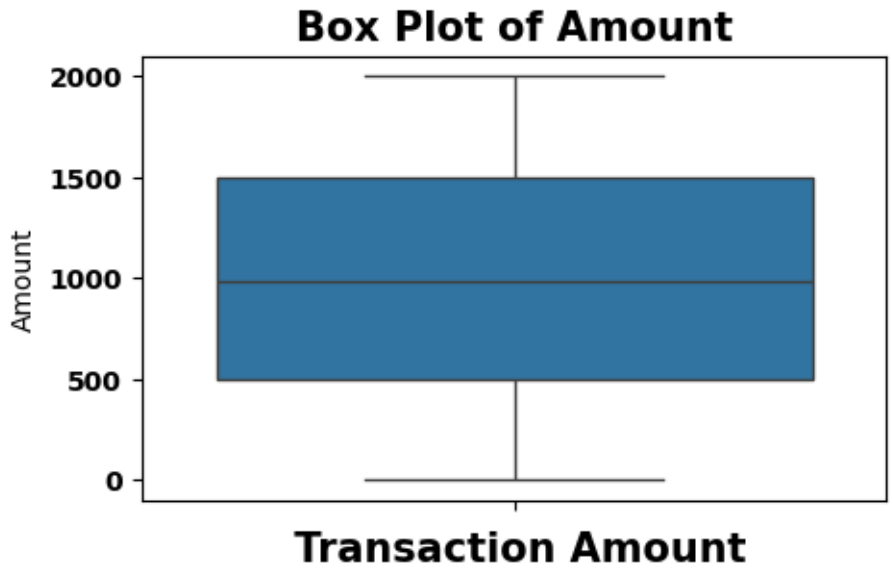


What we see: - Transaction amounts are spread almost uniformly between 0 and 2000. - No strong skew

Conclusion: - Transactions are evenly distributed across small and large amounts. - No dominance of only small-value transactions. - Good variation in transaction size.

2. Box plot for outliers

[32]:
```
plt.figure(figsize=(5,3))
sns.boxplot(df["Amount"],orient="horizontal")
plt.title("Box Plot of Amount",fontweight="bold",fontsize=15,color="black")
plt.xlabel(" Transaction Amount",fontweight="bold",fontsize=15,color="black")
plt.xticks(color="Black",fontweight="bold",fontsize=10)
plt.yticks(color="Black",fontweight="bold",fontsize=10)
plt.show()
```



Calculating Outliers

[9]:
```
q1=df["Amount"].quantile(0.25)
q3=df["Amount"].quantile(0.75)
iqr=q3-q1
```

[12]:
```
low=q1-1.5*iqr
high=q3+1.5*iqr
```

[13]:
```
((df["Amount"]<=low) | (df["Amount"]>=high)).sum()
```

[13]: `np.int64(0)`

What we see: - No extreme points outside whiskers. - Median is around the center of the data range.

Conclusion: - No major outliers in transaction amount. - Data looks clean and well-distributed. - No suspicious extremely high or low transactions.

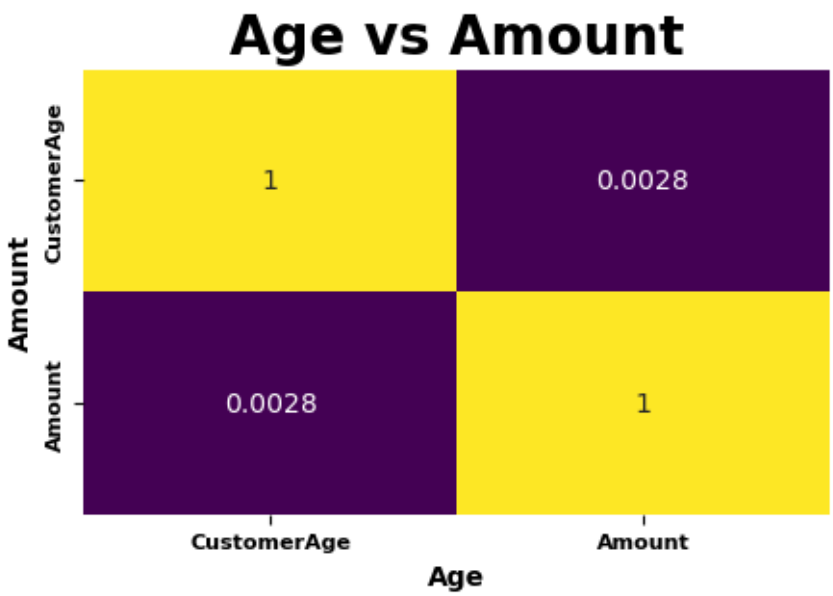
3. Correlation check (Age vs Amount)

[14]:
```
c=df[["CustomerAge","Amount"]].corr()
c
```

[14]:

	CustomerAge	Amount
CustomerAge	1.000000	0.002797
Amount	0.002797	1.000000

[16]:
```
plt.figure(figsize=(5,3))
sns.heatmap(c,alpha=1,annot=True,cmap="viridis",cbar=False)
plt.title("Age vs Amount",fontweight="bold",fontsize=20,color="black")
plt.xlabel("Age",fontweight="bold",fontsize=10,color="black")
plt.ylabel("Amount",fontweight="bold",fontsize=10,color="black")
plt.xticks(color="Black",fontweight="bold",fontsize=8)
plt.yticks(color="Black",fontweight="bold",fontsize=8)
plt.show()
```



Correlation coefficient ~0.0028

What this means: - Almost zero relationship.

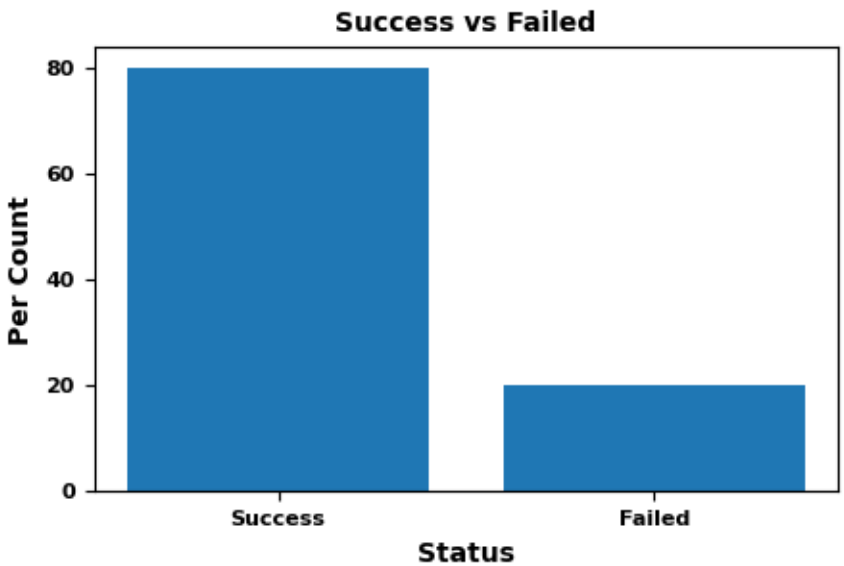
Conclusion: - Age does NOT influence transaction amount. - Young and older customers spend similarly. - No need to segment spending strategy by age (based on amount).

4. Success vs Failed proportion

[18]:
```
c=round(df["Status"].value_counts(normalize=True)*100,2)
c
```

[18]:
```
Status
Success    80.0
Failed     20.0
Name: proportion, dtype: float64
```

[20]:
```
plt.figure(figsize=(5,3))
plt.bar(c.index,c.values)
plt.title("Success vs Failed",fontweight="bold",fontsize=10,color="black")
plt.xlabel("Status",fontweight="bold",fontsize=10,color="black")
plt.ylabel("Per Count",fontweight="bold",fontsize=10,color="black")
plt.xticks(color="Black",fontweight="bold",fontsize=8)
plt.yticks(color="Black",fontweight="bold",fontsize=8)
plt.show()
```



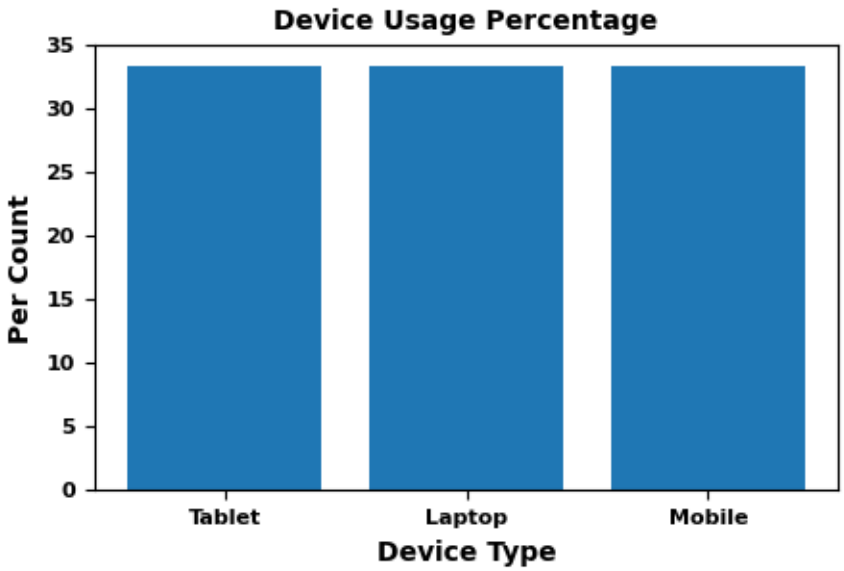
Conclusion: - System performance is good 80% success. - But 20% failure is still significant. - You may investigate failure reasons network, bank issue, device, etc..

5. Device usage %

[21]:
```
d=round(df["DeviceType"].value_counts(normalize=True)*100,2)
d
```

[21]:
```
DeviceType
Tablet    33.34
Laptop    33.33
Mobile    33.33
Name: proportion, dtype: float64
```

[23]:
```
plt.figure(figsize=(5,3))
plt.bar(d.index,d.values)
plt.title("Device Usage Percentage",fontweight="bold",fontsize=10,color="black")
plt.xlabel("Device Type",fontweight="bold",fontsize=10,color="black")
plt.ylabel("Per Count",fontweight="bold",fontsize=10,color="black")
plt.xticks(color="Black",fontweight="bold",fontsize=8)
plt.yticks(color="Black",fontweight="bold",fontsize=8)
plt.show()
```



Conclusion: - Device usage is perfectly balanced. - No single dominant device. - Platform is equally used across devices. - System optimization should be maintained for all devices equally.

[24]:
```
c=df[["CustomerAge","Amount"]].corr()
c
```

[24]:

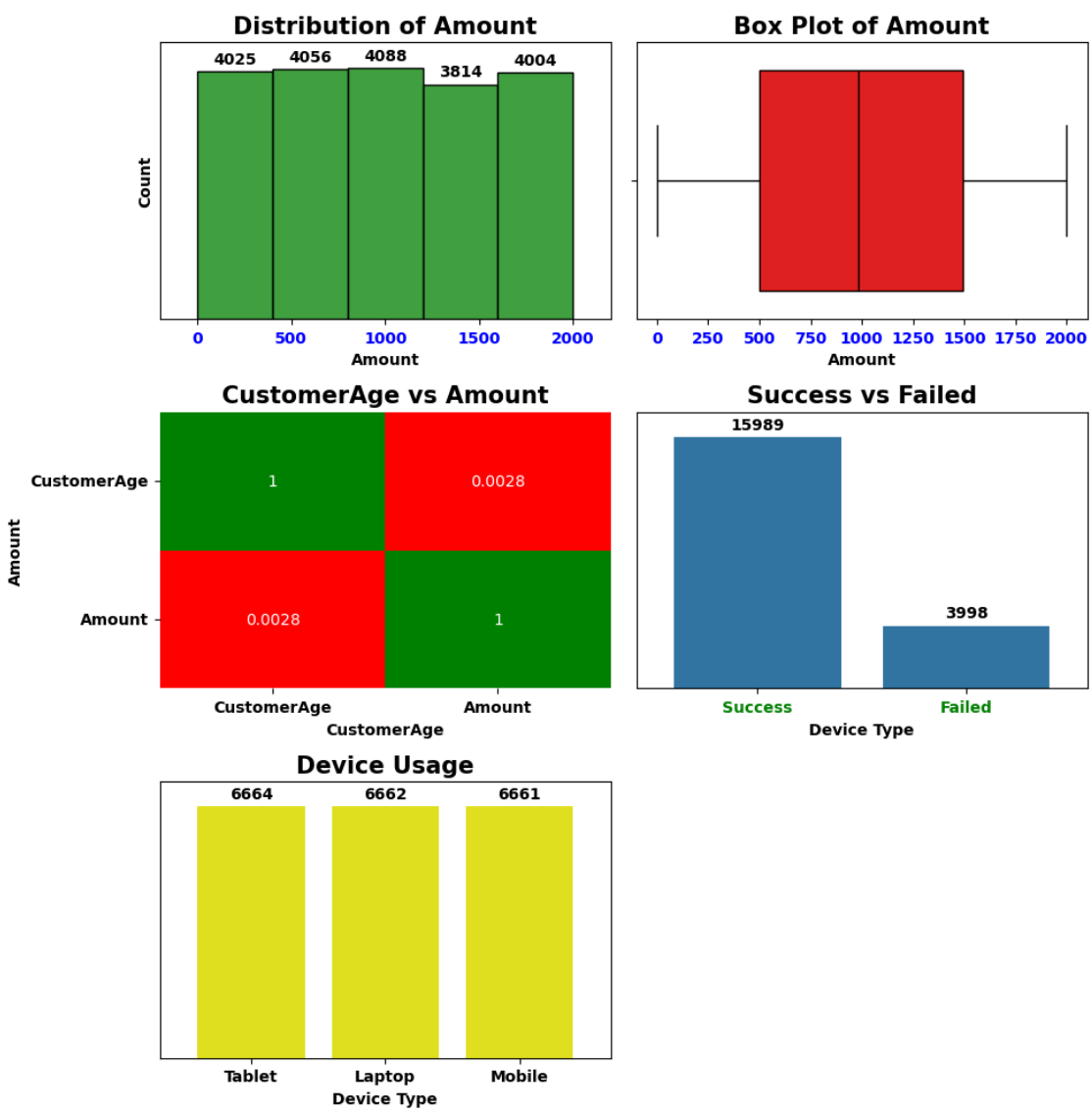
	CustomerAge	Amount
CustomerAge	1.000000	0.002797
Amount	0.002797	1.000000

1 Combine All Charts

[25]:
```
from numpy import ndarray
fig,ax=plt.subplots(nrows=3,ncols=2,figsize=(10,10))
plt.subplots_adjust(wspace=2,hspace=3)

sns.histplot(df["Amount"],bins=5,ax=ax[0,0],color="green")
ax[0,0].set_title("Distribution of Amount",fontweight="bold",fontsize=15)
```

```
ax[0,0].set_xlabel("Amount",fontweight="bold")
ax[0,0].set_ylabel("Count",fontweight="bold")
ax[0,0].set_yticks([])
ax[0,0].tick_params(labelcolor="blue")
for i in ax[0,0].get_xticklabels():
    i.set_fontweight("bold")
for i in ax[0,0].get_yticklabels():
    i.set_fontweight("bold")
for i in ax[0,0].containers:
    ax[0,0].bar_label(i,fontweight="bold",padding=3)
ax[0,0].margins(0.1)



sns.boxplot(x=df["Amount"],ax=ax[0,1],orient="y",linecolor="black",color="red")
ax[0,1].tick_params(labelcolor="blue")
ax[0,1].set_title("Box Plot of Amount",fontweight="bold",fontsize=15)
ax[0,1].set_xlabel("Amount",fontweight="bold")
for i in ax[0,1].get_xticklabels():
    i.set_fontweight("bold")



sns.heatmap(c,ax=ax[1,0],cmap=["red","green"],cbar=False,annot=True)
ax[1,0].set_xlabel("CustomerAge",fontweight="bold")
ax[1,0].set_ylabel("Amount",fontweight="bold")
ax[1,0].set_title("CustomerAge vs Amount",fontweight="bold",fontsize=15)
for i in ax[1,0].get_xticklabels():
    i.set_fontweight("bold")
for i in ax[1,0].get_yticklabels():
    i.set_fontweight("bold")



sns.barplot(x=df["Status"].value_counts().index,y=df["Status"].value_counts().
 ↪values,ax=ax[1,1])
ax[1,1].tick_params(labelcolor="green")
ax[1,1].set_title("Success vs Failed",fontweight="bold",fontsize=15)
for i in ax[1,1].get_xticklabels():
    i.set_fontweight("bold")
ax[1,1].set_yticks([])
for i in ax[1,1].containers:
    ax[1,1].bar_label(i,fontweight="bold",padding=3)
ax[1,1].margins(0.1)
ax[1,1].set_xlabel("Device Type",fontweight="bold")
```

```
sns.barplot(x=df["DeviceType"].value_counts().index,y=df["DeviceType"].
 ↪value_counts().values,ax=ax[2,0],color="yellow")
ax[2,0].set_title("Device Usage",fontweight="bold",fontsize=15)
for i in ax[2,0].containers:
    ax[2,0].bar_label(i,fontweight="bold",padding=3)
ax[2,0].set_yticks([])
ax[2,0].margins(0.1)
for i in ax[2,0].get_xticklabels():
    i.set_fontweight("bold")
ax[2,0].set_xlabel("Device Type",fontweight="bold")


fig.delaxes(ax[2,1])
plt.tight_layout()
plt.show()
```



[ ]:

```

```